# Benjamin Zhang

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## Education

**University of Southern California** — Los Angeles, CA *Fall 2022–Spring 2027 (expected)*

- PhD candidate in the department of Physics and Astronomy.
- Cumulative GPA: 3.9

**University of California, Berkeley** — Berkeley, CA *Spring 2017–Spring 2021*

- B.A. in Astrophysics and Computer Science, *summa cum laude*
- Cumulative GPA: 3.98

## Honors, Awards & Fellowships

**W. M. Keck Foundation Scholar-Fellow Bridge Initiative Fellowship** *1/2026*

## Publications

A continuously updated publication list can be found on ORCID.
† indicates equal-contribution authorship (names in alphabetical order).

1. Kris Pardo, Eleanor Stuart†, **Benjamin Zhang**† et al. *Searching for Gravitational Waves with the Kepler Space Telescope: First Limits on Continuous Microhertz Waves*. To be submitted Nov. 2026 to *Physical Review Letters*.

2. **Benjamin Zhang**, Eleanor Stuart, Kris Pardo et al. *Searching for Gravitational Waves with the Kepler Space Telescope: Statistical Techniques and Likelihood Construction*. To be submitted Nov. 2026 to *Physical Review D*.

3. Eleanor Stuart, **Benjamin Zhang**, Kris Pardo et al. *Searching for Gravitational Waves with the Kepler Space Telescope: Astrometric Modeling and Systematics Mitigation*. To be submitted Nov. 2026 to *Physical Review D*.

4. **Benjamin Zhang**, Kris Pardo et al. *A compact time-domain reduced order quadrature for gravitational wave inspirals with non-stationary noise*. Submitted Oct. 2026 to *Physical Review Letters*. arXiv:2609.14130

5. **Benjamin Zhang**, Khee-Gan Lee et al. *Cross-correlations between the CLAMATO Lyman-alpha forest and galaxies within the COSMOS field*. *Monthly Notices of the Royal Astronomical Society*, vol. 548, iss. 4, Jun. 2026. doi:10.1093/mnras/stag786

6. **Benjamin Zhang**, Kris Pardo et al. *Fast Bayesian method for coherent gravitational wave searches with relative astrometry*. *Physical Review D*, 112, 042002, Aug. 2025. doi:10.1103/qzys-t232

7. **Benjamin Zhang**, Khee-Gan Lee et al. *Alignments between Galaxies and the Cosmic Web at z 1-2 in the IllustrisTNG Simulations*. *The Astrophysical Journal*, vol. 954, no. 1, Aug. 2023. doi:10.3847/1538-4357/ace695.

8. Benjamin Horowitz, **Benjamin Zhang** et al. *TARDIS Paper II: Synergistic Density Reconstruction from Lyman-alpha Forest and Spectroscopic Galaxy Surveys with Applications to Protoclusters and the Cosmic Web*. *The Astrophysical Journal*, vol. 906, no. 2, Jan. 2021. doi:10.3847/1538-4357/abca35.

## Research Experience

**University of Southern California** — Los Angeles, CA *Fall 2022–Present*

- Working with Professor Kris Pardo on measuring gravitational waves with relative stellar astrometry, using data from the *Kepler Space Telescope*.
- Developed scalable computational pipeline for reduction of systematics on terabyte-level archival *Kepler* astrometric data.
- Extended computational methods from ground-based gravitational wave detectors to efficiently search for complex, inspiralling waveforms in astrometric data.
- Obtained first mock data-based constraints on detection of continuous gravitational waves with astrometry, using novel computational method for fast Bayesian searches.

**Center for Computational Astrophysics** — New York, NY *Summer 2026*

- In-person visitor at Flatiron Institute's Center for Computational Astrophysics from July 2026 to August 2026, hosted by Professor Will M. Farr.
- *Ongoing:* working to exploit the statistical geometry of gravitational waveforms to efficiently test for the possible presence of beyond-GR waveform effects.

**Kavli IPMU** — Chiba, Japan *Summer 2020–Summer 2025*

- In-person visiting researcher at Kavli IPMU from Oct 2023-Dec 2023/Dec 2024-Jan 2025, hosted by Professor Khee-Gan Lee.
- Studied the cross-correlation between galaxies and surrounding large-scale density (through absorption of the Lyman-alpha forest) for galaxies within the COSMOS Lyman-Alpha Mapping And Tomography Observations (CLAMATO) survey.
- Obtained constraints on the dark matter halo mass-galaxy stellar mass relation at high-redshifts using this cross-correlation.
- In-person visiting researcher at Kavli IPMU from Fall 2021 to Fall 2022, hosted by Professor Khee-Gan Lee.
- Researched strength of galaxy-cosmic web intrinsic alignments at high redshifts for upcoming Subaru Prime Focus Spectrograph galaxy spectroscopy survey, with Professor Khee-Gan Lee.
- Applied redshift space distortion and cosmic web reconstruction from mock spectrographic surveys to cosmological simulation & associated galaxies, to quantify how alignment signal changes with observational effects.

**University of California, Berkeley** — Berkeley, CA *Spring 2019–Summer 2020*

- Worked with Dr. Benjamin Horowitz on universe large-scale structure reconstruction from Lyman-alpha absorption features in high-redshift quasar and galaxy spectra (Lyman-alpha forest tomography).
- Created code to quantify correlated error from high-redshift quasar/galaxy spectrum continuua being misestimated, in Python. Modified Dr. Horowitz's Lyman-alpha tomography reconstruction code, TARDIS, to use the correlated error, improving reconstruction accuracy.
- Wrote Tensorflow code for new major version of TARDIS, and tested its numerical stability.
- Developed script to produce 3D movies of TARDIS converging on final structure reconstructions, using Python and yt library.

## Presentations

**DiRAC Seminar** — University of Washington, Seattle *10/2026*

- Talk: *Searching for continuous gravitational waves from supermassive black hole binaries using relative stellar astrometry*

**LIGO Seminar** — California Institute of Technology *10/2026*

- Invited talk: *Gravitational wave detection with relative stellar astrometry*

**International Pulsar Timing Array Science Meeting 2025** — California Institute of Technology *6/2025*

- Contributed talk: *Fast coherent GW searches for astrometry & PTAs using LIGO methods*

**APS Joint March Meeting and April Meeting/Global Physics Summit 2025** — Anaheim, CA *3/2025*

- Contributed talk: *Searching for coherent gravitational waves with relative stellar astrometry*

**International Pulsar Timing Array Science Meeting 2024** — Sexten Center of Astrophysics *6/2024*

- Poster: *Searching for coherent gravitational waves with relative stellar astrometry*

**KITP Conference: The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time** — University of California, Santa Barbara *2/2023*

- Contributed talk: *Intrinsic Alignments Between Galaxies and the Cosmic Web at $z \sim$ 1-2 in the IllustrisTNG Simulations*

**YITP Molecule-type Workshop on Galaxy shape statistics and Cosmology** — Kyoto University *11/2021*

- Contributed talk: *Forecasting $z \sim 2$ direct galaxy-cosmic web alignment for Subaru-PFS*

**Prime Focus Spectrograph Twelveth Collaboration Meeting** — Virtual *03/2021*

- Virtual poster: *Forecasting High-z Galaxy-Cosmic Web Alignments for the Subaru Prime Focus Spectrograph Survey*

**237th meeting of the American Astronomical Society** — Virtual *01/2021*

- Virtual poster: *Forecasting Galaxy-Cosmic Web Alignments for High-z Galaxies*

## Teaching Experience

**Teaching Assistant** — PHYS 430: General Relativity & Gravitation *Fall 2023*

## Service

- NASA ROSES grant proposal review panel member

## Skills

- Experienced with high-performance computing systems, Slurm, Docker, data pipeline orchestration.
- Extensive experience with Python scientific computing libraries: JAX, JAXNS, Numpy/Scipy/Matplotlib/Astropy. Experience with libraries: Numpyro, Blackjax.
- Proficient in Japanese: certified JLPT N2 (CEFR Level B2)